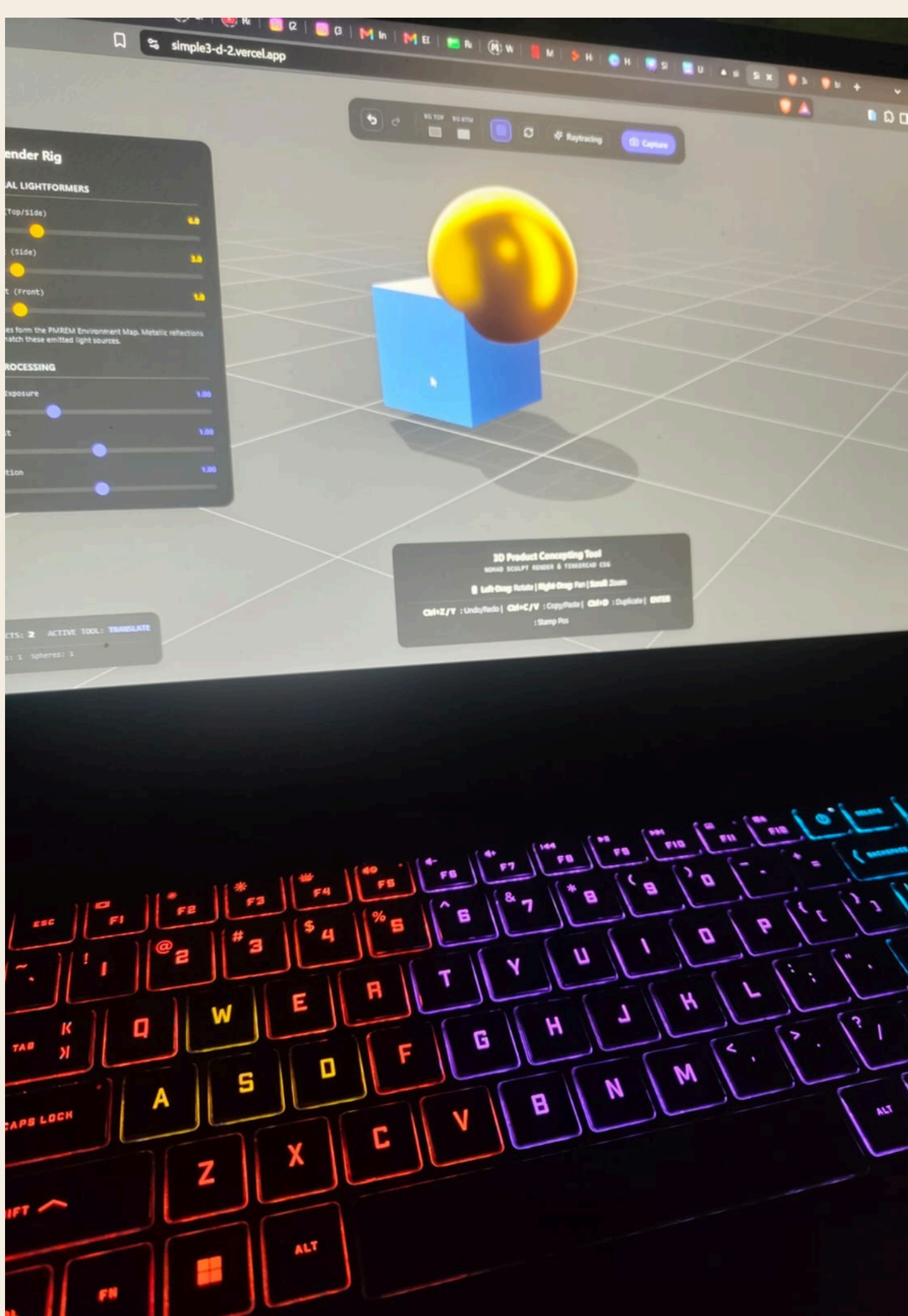




SIMPLE 3D · MID-SEMESTER PROJECT PITCH

SIMPLE 3D: HIGH-END 3D, MADE SIMPLE

Core Objective: POC

- Prove high-fidelity 3D rendering in the browser
- Combine professional visuals with beginner-friendly controls
- Target: Blender/Nomad-level fidelity + Tinkercad-level ease

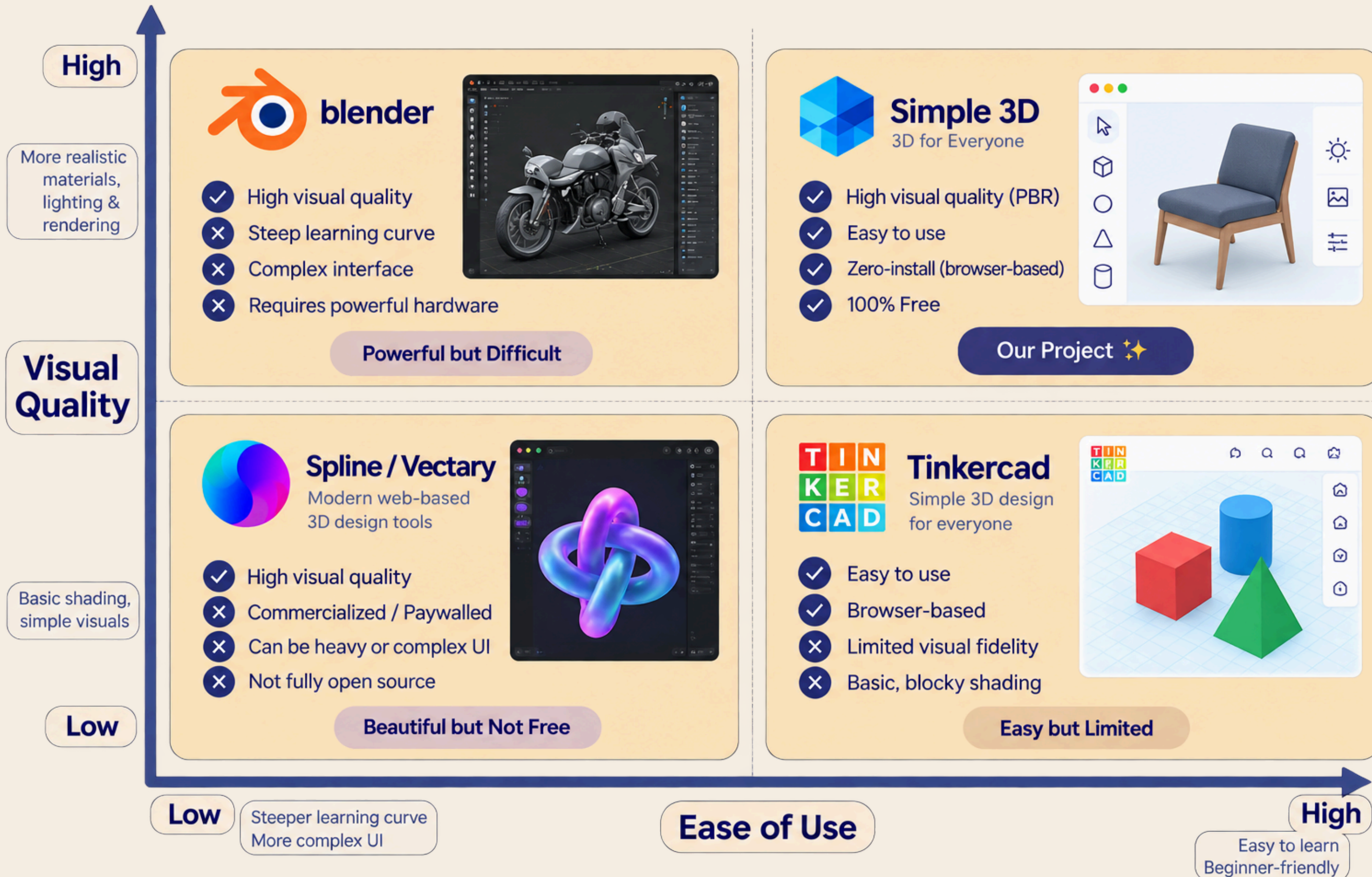
What is Simple 3D?

- Zero-install, browser-based 3D workspace
- Designed for easy and accessible 3D creation

Engineering Feat

- Real-time, client-side rendering
- Stable and performant web-based 3D engine
- Professional-grade graphics without installation

THE PROBLEM: A GAP IN 3D CREATION



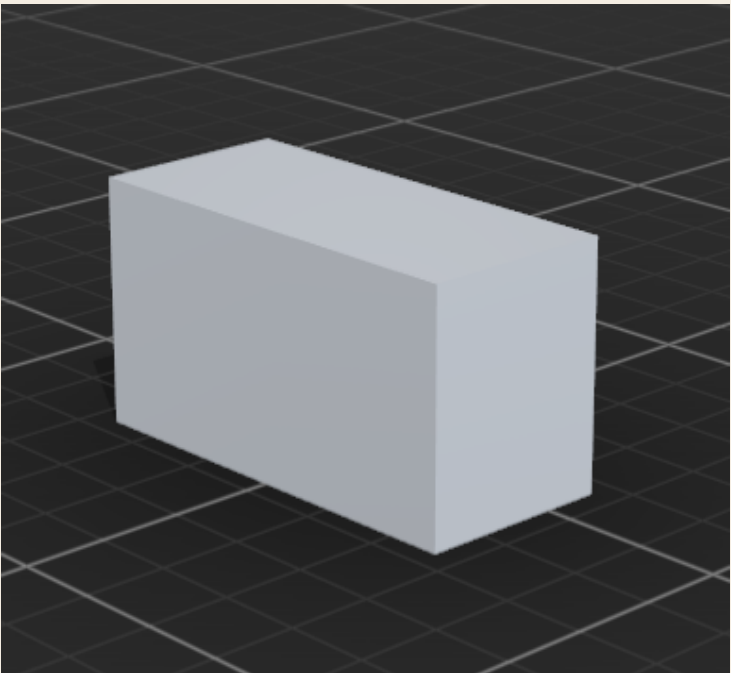
VISUAL PROOF: HIGH-END 3D IN THE BROWSER



The Technical Leap

- PBR materials:
- Metalness • Roughness • Clearcoat
- Transmission
- Real-time reflections & lighting
- Lighting separated from the background
- Runs directly in the web browser

High-fidelity studio-quality visuals: instantly, client-side.



PBR MATERIAL

Base Color ■

Metalness 0.30

Roughness 0.57

Clearcoat 0.38

Transmission (Glass) 0.64

IOR 1.81

Studio Render Rig

PROCEDURAL LIGHTFORMERS

Key Light (Top/Side) 6.0

Edge Light (Side) 3.0

Fill Light (Front) 1.0

These planes form the PMREM Environment Map. Metallic reflections perfectly match these emitted light sources.

POST-PROCESSING

Camera Exposure 1.00

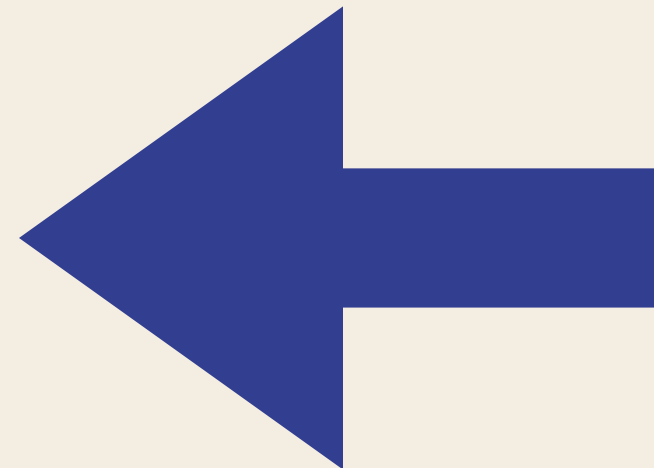
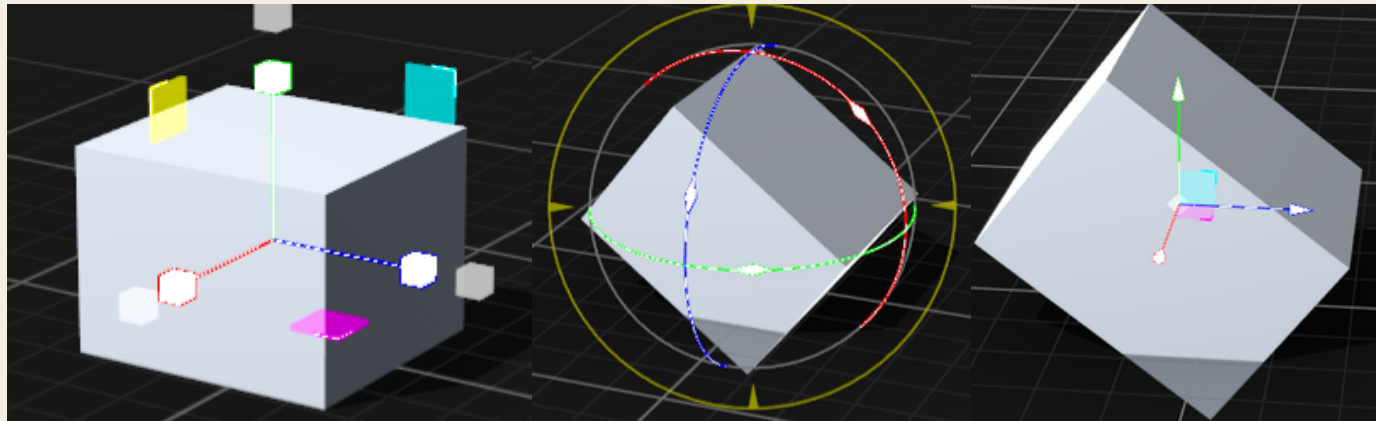
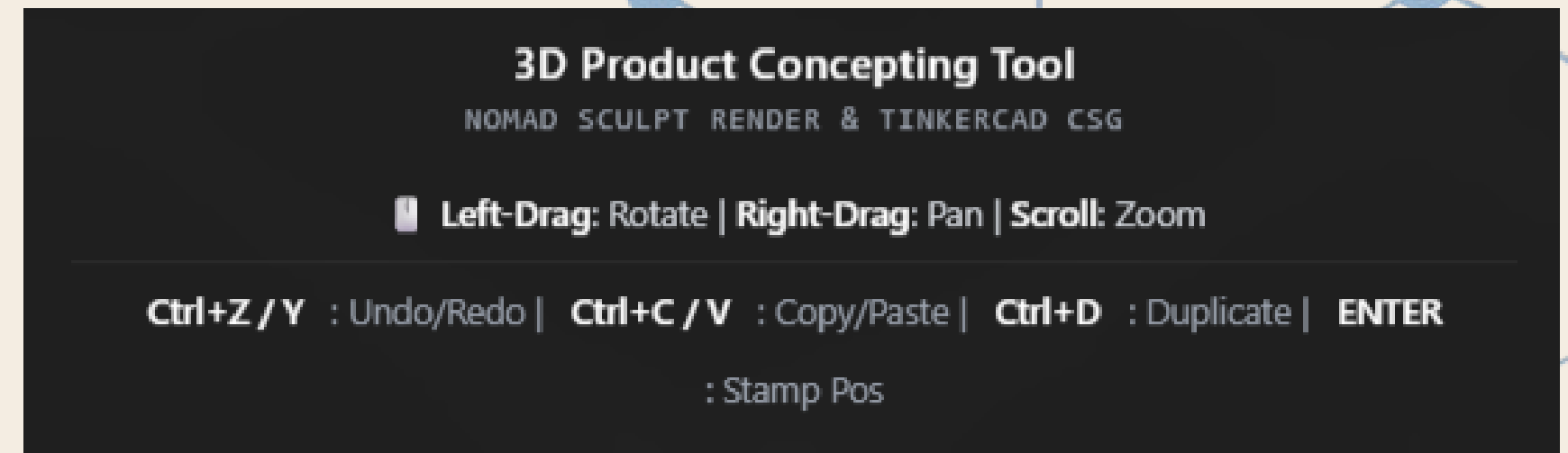
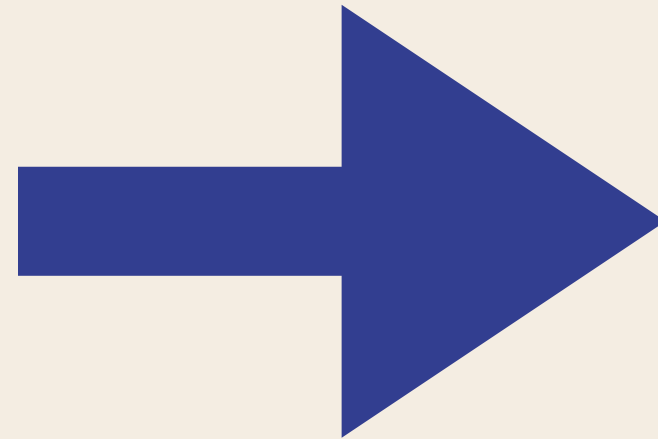
Contrast 1.00

Saturation 1.00

INTERFACE & EXPERIENCE

Effortless Interaction:

- Zero prior 3D knowledge required
- Intuitive drag, scale & rotate controls

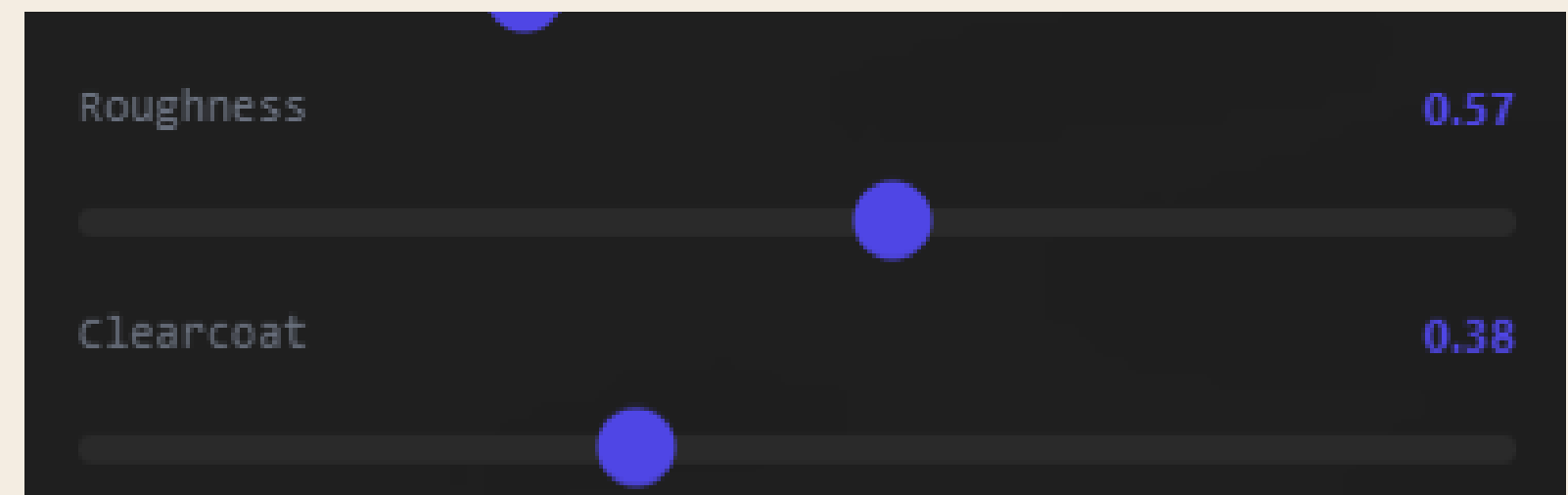
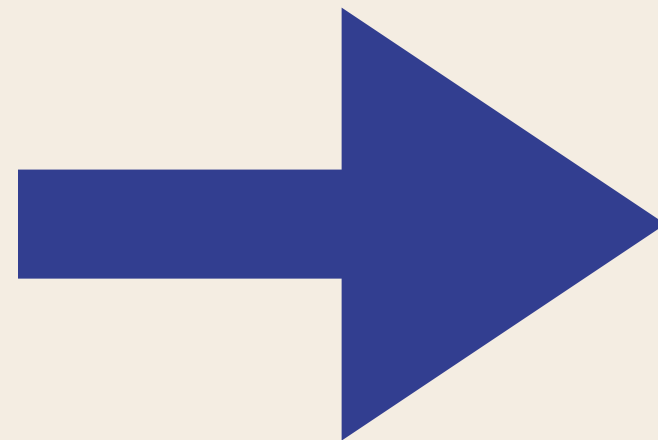


Smooth Transformations:

- Continuous state synchronization with onObjectChange
- Smooth, responsive TransformControls
- No snapping or interaction glitches

Minimalist Controls:

- Simple sliders for PBR properties
- Create shiny metal, glass & frosted materials with ease



FEATURE DEVELOPMENT ROADMAP

Currently Developing

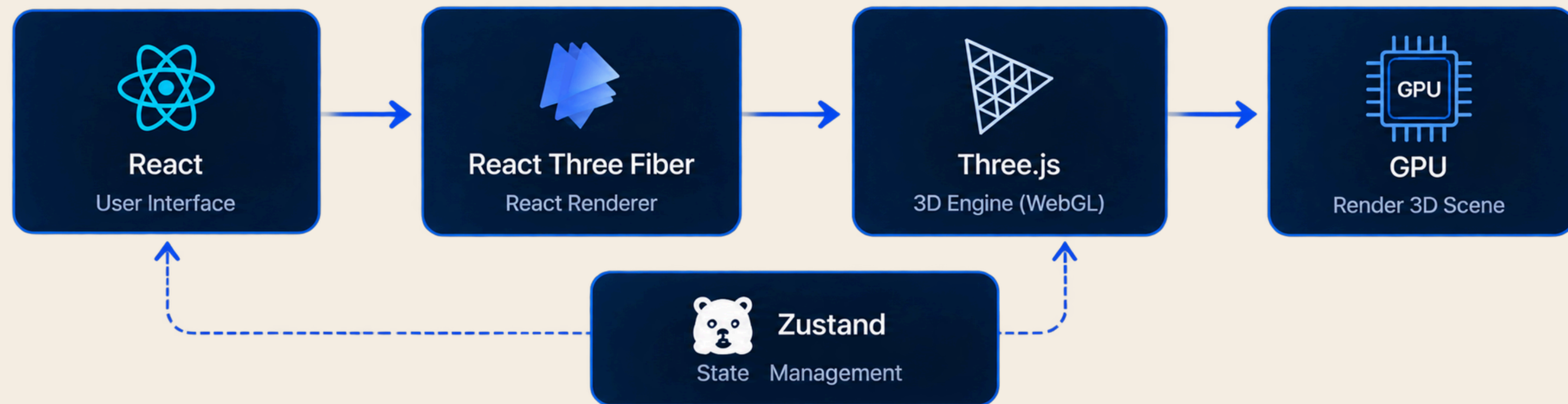
- PBR Material System — Real-time metalness, roughness, clearcoat & transmission
- Dynamic Studio Lighting — Key, Edge & Fill with procedural backgrounds
- Smooth Transformations — Real-time, responsive mesh interaction

Next Development Goals

- CSG Booleans — Real-time cut & combine operations
- Edge Filleting — Client-side geometry refinement
- GLTF / OBJ Export — For 3D printing & external workflows

Goal: Continuously improve performance, stability & visual fidelity.

PROJECT ARCHITECTURE: FROM UI TO GPU



ENGINEERING CHALLENGE

- React's declarative UI ↔ Three.js's imperative render loop
- Continuous synchronization between UI state & GPU transforms
- Optimized to maintain smooth real-time rendering
- Minimizes unnecessary React re-renders

METHODOLOGY: AGILE DEVELOPMENT

PHASE 1 — FOUNDATION

WebGL + React Integration

Set up the core 3D rendering environment.

PHASE 3 — CURRENT

PBR + Lighting + Deployment

Refining rendering, lighting, and Vercel CI/CD

PHASE 2 — STATE SYNC

React + Three.js + Zustand

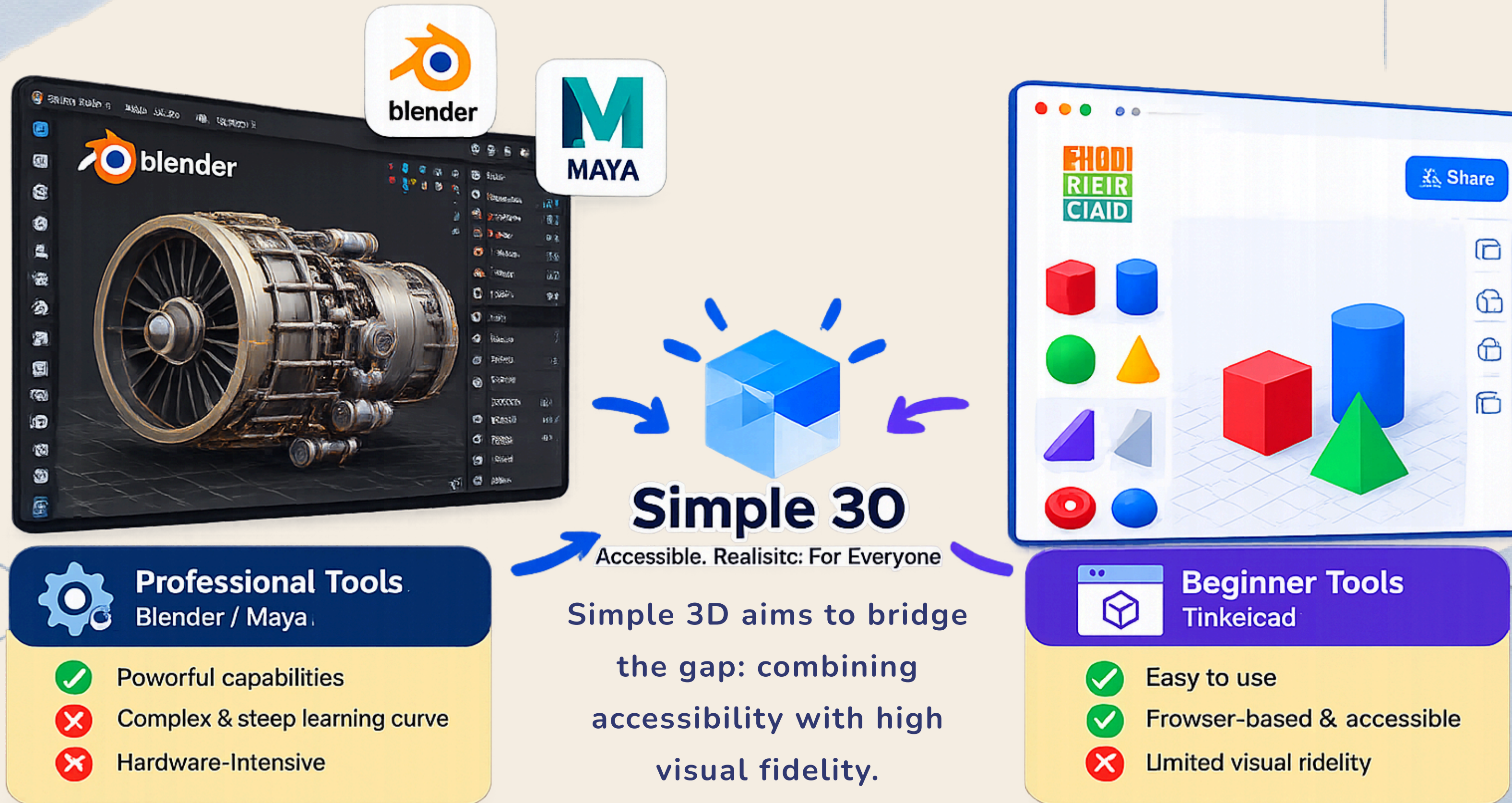
Established reliable real-time state synchronization.

PHASE 4 — NEXT

CSG + File I/O

Developing advanced geometry operations and GLTF/OBJ export.

THE PROBLEM: A GAP IN 3D CREATION



COSTS & SUSTAINABILITY

01



Client-Side Computing

- ✓ Rendering runs on the user's GPU
- ✓ CSG computation runs locally
- ✓ Minimizes server-side processing

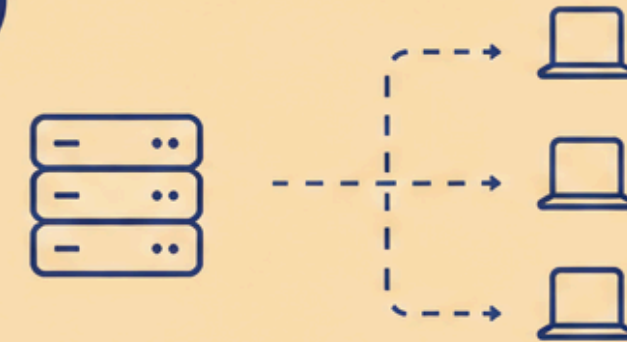
02



Low-Cost Deployment

- ✓ Hosted through Vercel
- ✓ Lightweight backend requirements
- ✓ Suitable for a student-scale POC

03

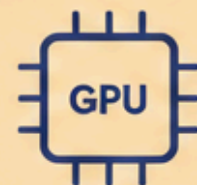


Scalable Architecture

- ✓ More computation happens on the user's device
- ✓ Server load is less dependent on rendering complexity
- ✓ Supports growth without proportional rendering infrastructure



User's Device
(Browser)



WebGL / GPU
(Local Processing)



Minimal Server
Dependency

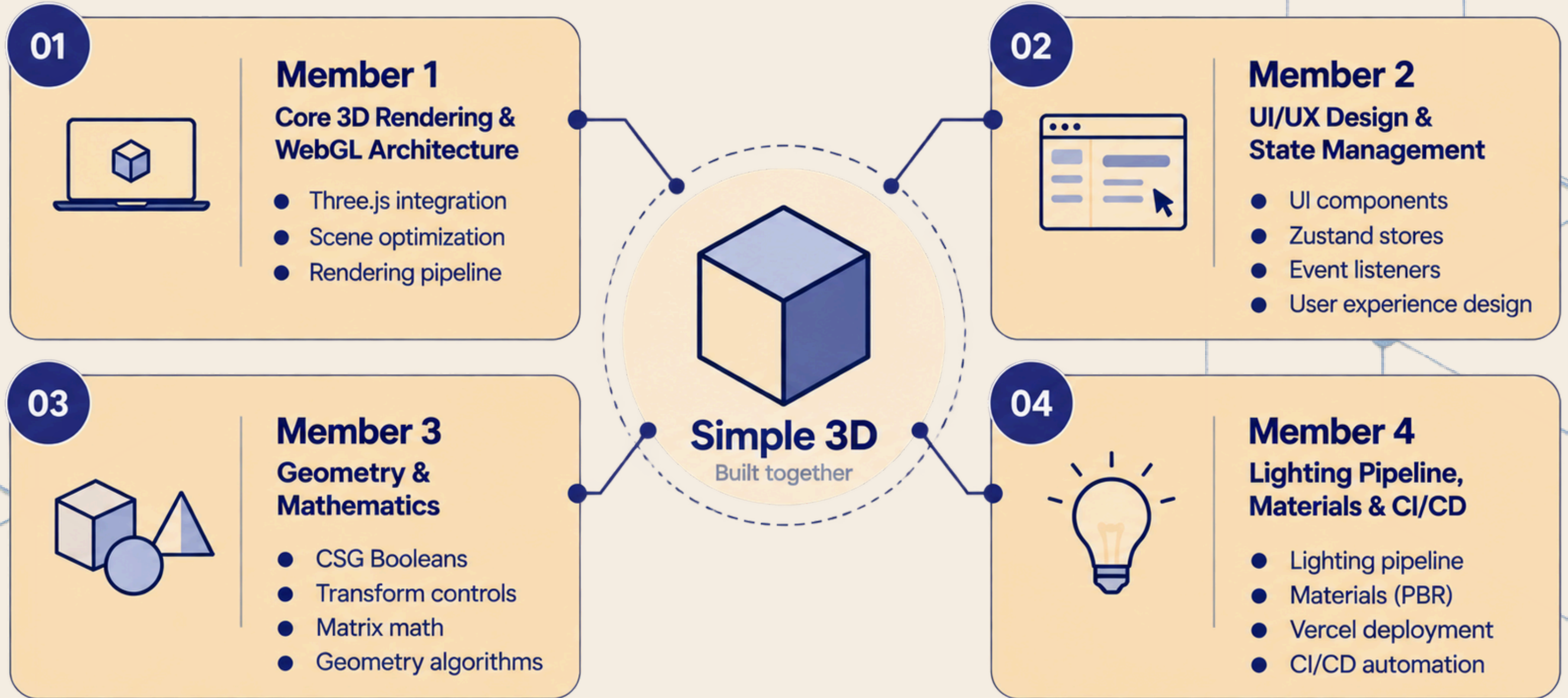


Lower
Infrastructure Cost



Compute where the work happens — on the user's device.

RESPONSIBILITY DISTRIBUTION

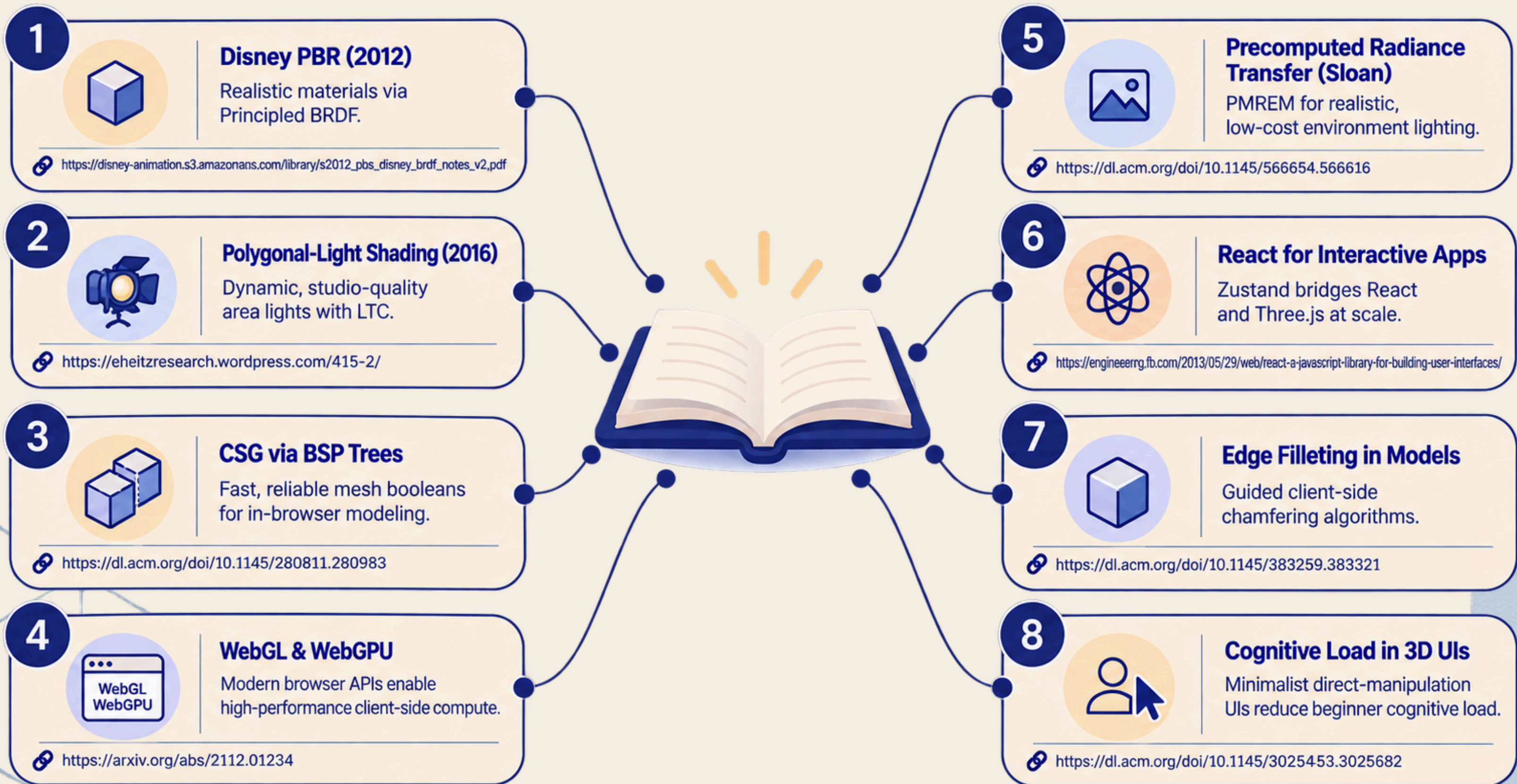


Different strengths. One cohesive product.



Build • Design • Compute • Together

LITERATURE REVIEW & RESEARCH



WHAT COMES NEXT

Simple 3D shows that ease of use and visual quality can coexist combining high-end PBR rendering with a smooth, intuitive browser experience.

- **What's Next**

- Smart Dimensions — Precise measurements & modeling
- CSG Booleans — Stable cut & combine operations
- Edge Filleting — Smoother, refined geometry
- File Export — GLTF / OBJ workflows

- **Final Thought**

- Democratizing high-quality 3D for the next generation of designers, hobbyists & engineer

THANK YOU!